Predictive Customer Churn Analytics

Telecom retention

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PROJECT SCOPE

One selected project, built in IBM SPSS Modeler

Selected project:PredictiveCustomerChurnAnalyticsfortelecomretention.

- Tool: IBMSPSSModeler
- Dataset understanding, then data cleaning and preparation
- Data transformation: Churn set as the flag target, predictors as inputs
- Model building (C5.0) and model evaluation on a train–test split
- Interpretation of results for retention action

Additional features: none were created. The model uses the telecom_churn fields as provided.

Churn prediction turns retention into an early-warning system

The model shifts retention work upstream, before attrition becomes irreversible.

Business problem

Identify customers likely to leave before attrition becomes irreversible.

Business value

Focus retention effort where intervention can protect existing relationships.

Project objective

Build an interpretable binary classifier in IBM SPSS Modeler.

The dataset combines account, usage, and service signals

The analytical question starts with a clear separation between what describes the customer and what the model must explain.

ACCOUNT STRUCTURE

AccountWeeks, Contract, AccountLength, DataPlan

USAGE Behavior and charges

DayMins, DayCalls, MonthlyCharges, DataUsage, OverageFee, RoamMins

SERVICE
Support pressure

CustServCalls

TARGET
Churn

A binary flag for retained versus churned customers.

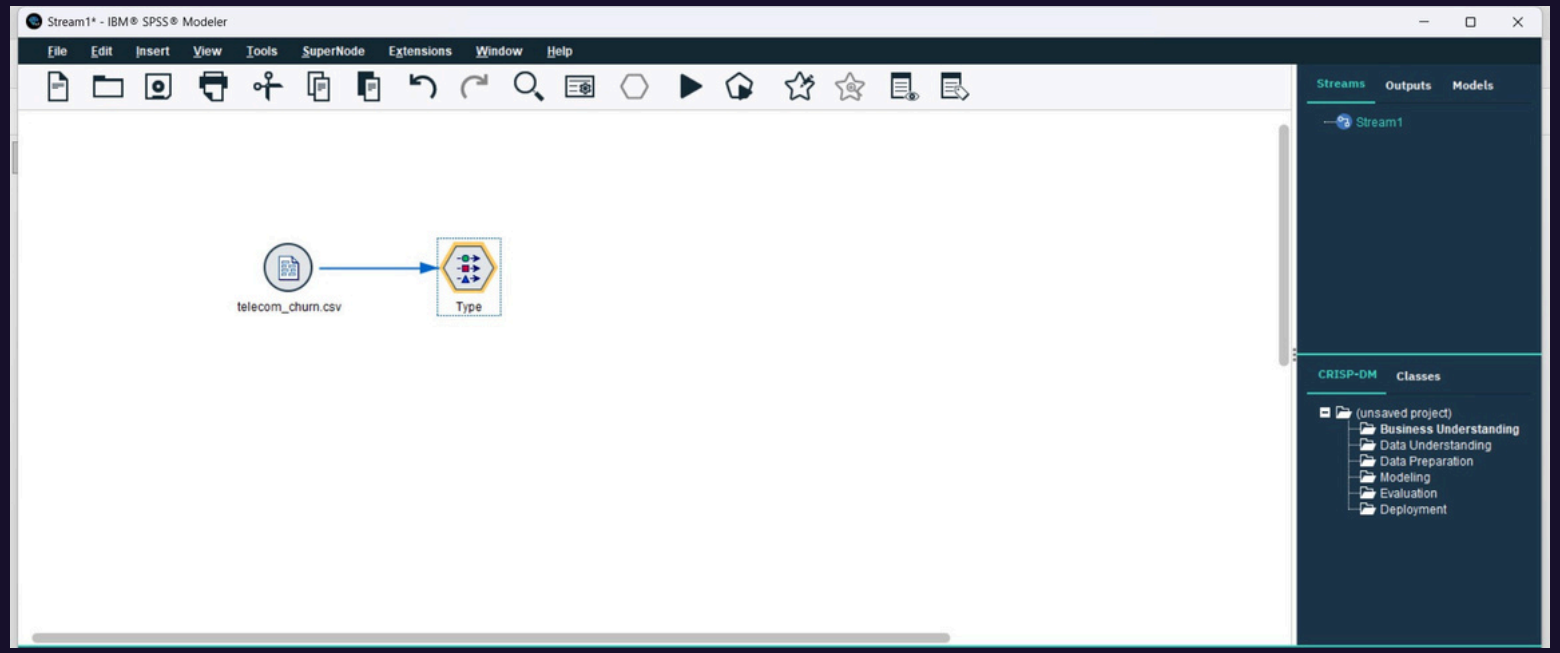
The stream converts raw records into model-ready fields

Import
Import telecom_churn.csv with the Var. File node.

Assign
Use Type to assign Churn as Target and predictors as Input.

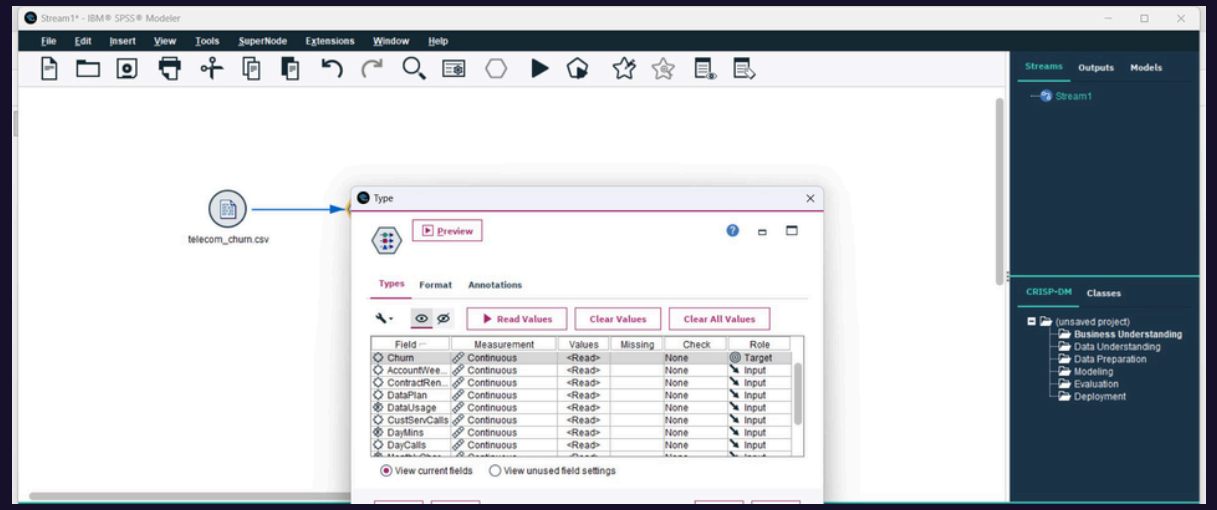
Convert
Convert Churn to Flag/Nominal for classification.

Partition
Partition the records before training and testing.



Field roles make the analytical question explicit

Churn is the outcome to explain; account, usage, and service variables are the signals used to explain it.



Outcome

Churn

Signals

Account and usage f ields

Context

Service variables

05 — VALIDATION DESIGN

The shown run uses a balanced train–test split

TRAINING

50%

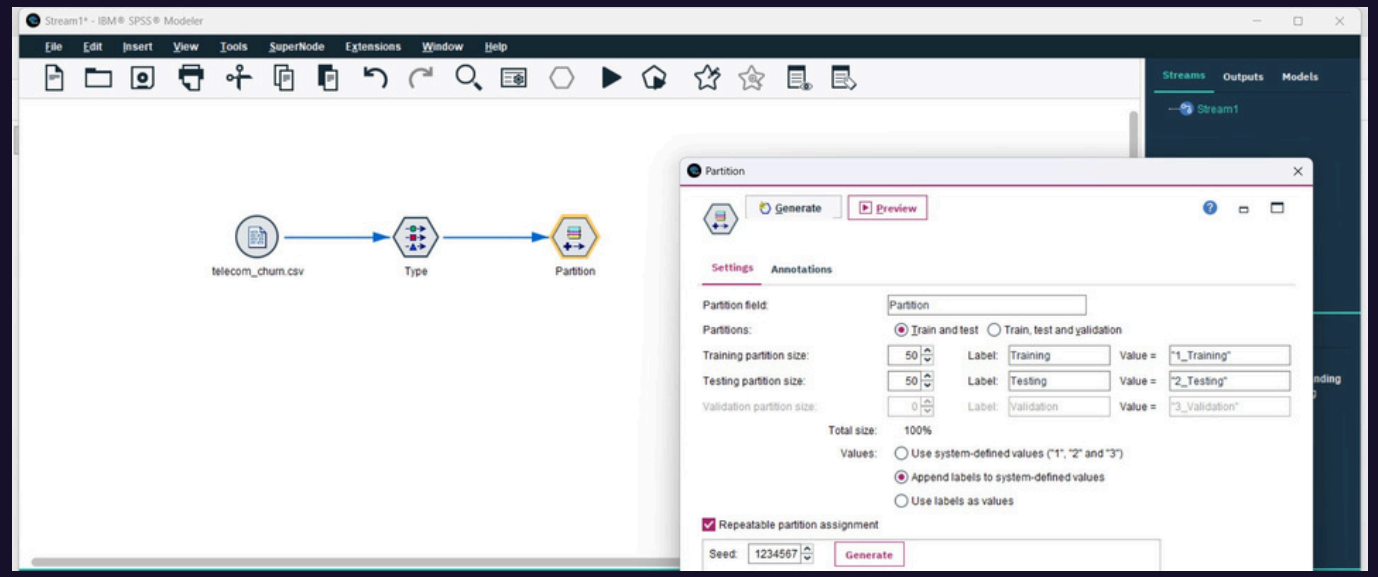
of records

TESTING

50%

of records

Purpose: assess performance on unseen data and reduce the risk of overfitting.



C5.0 turns churn risk into interpretable rules

The model keeps the reasoning visible from partition to prediction.

Partition

Feeds the model with the prepared records.

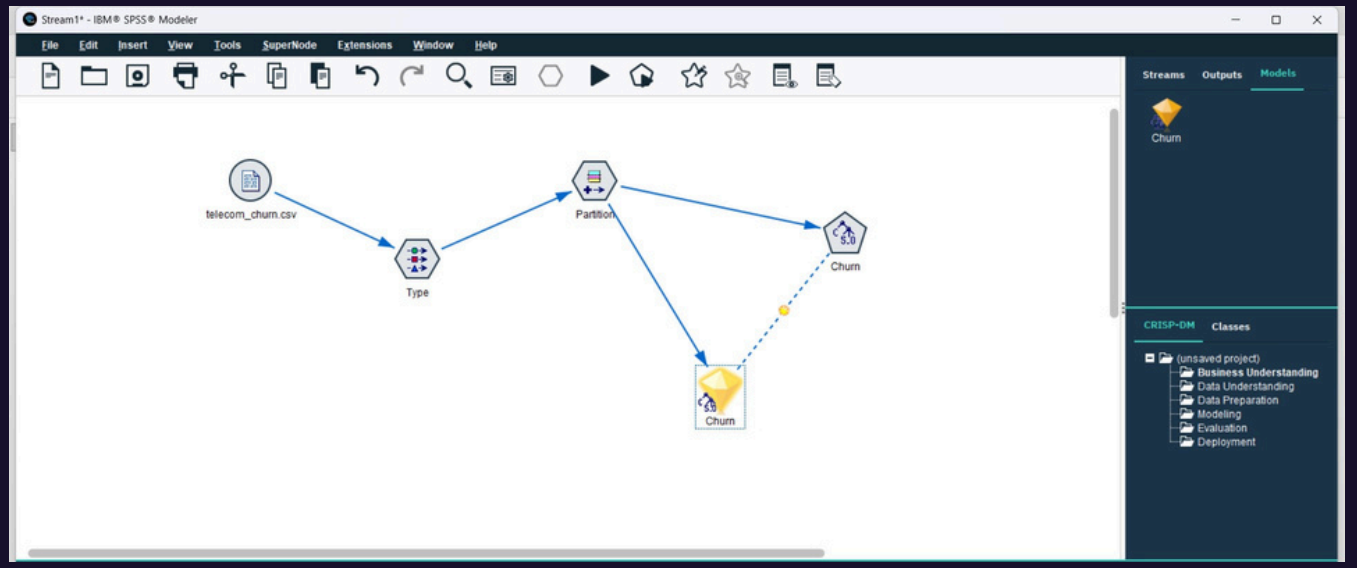
C5.0

Builds a decision tree for binary classification .

Model nugget

Produces a reusable churn model.

Why it fits: binary classification, transparent logic, and business-readable rules.



The model reaches 92.65% accuracy on the shown testing partition

TESTING

92.65%

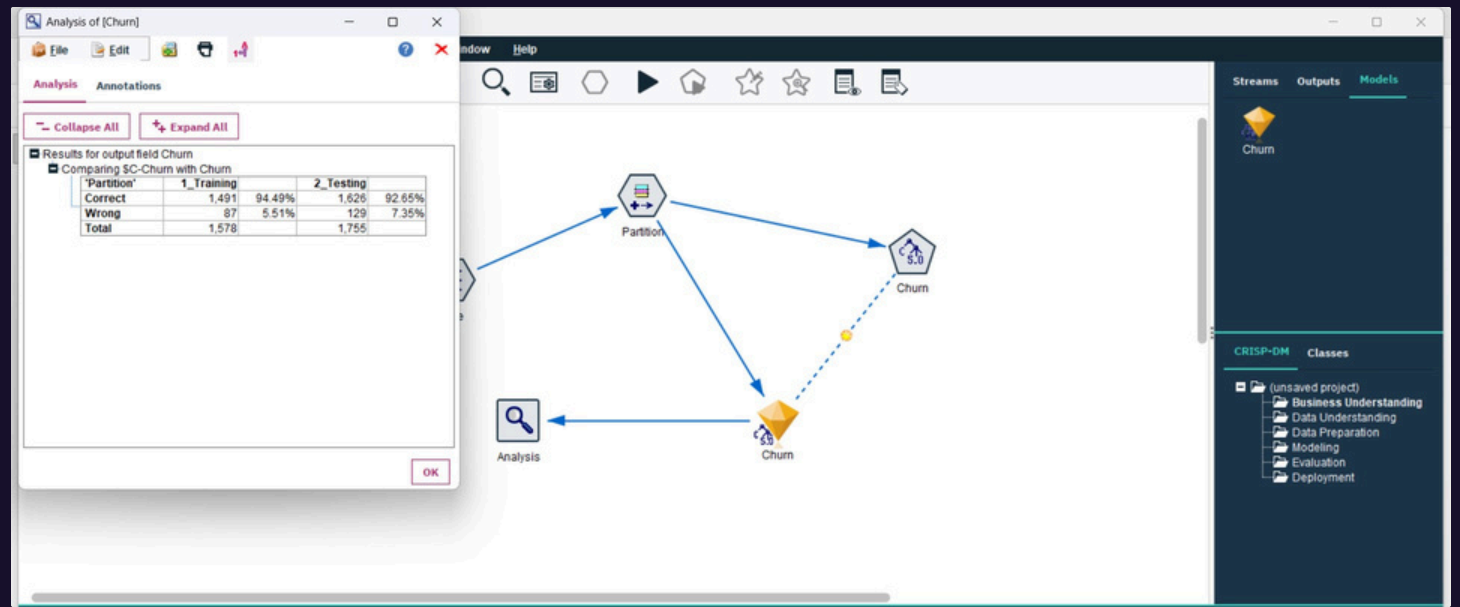
1,626 correct out of 1,755

TRAINING

94.49%

1,491 correct out of 1,578

Performanceremainsstrongonunseenrecords,with a modest train–test gap.



Accuracy is the starting point, not the retention strategy



Intervene

Translate high-risk rules into targeted offers, service recovery, or plan changes.



Review misses

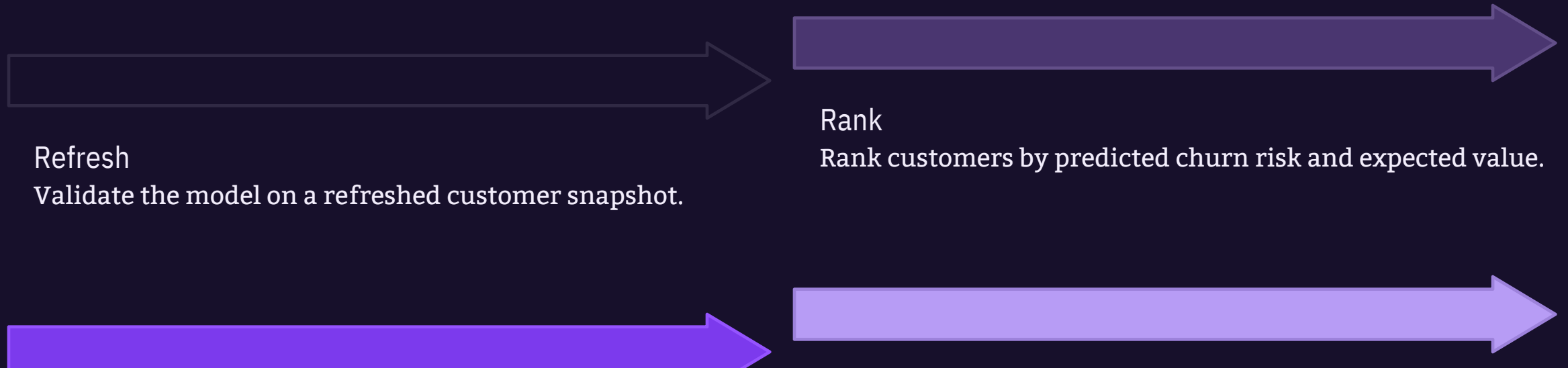
Review false negatives separately because missed churn can be costly.



Compare segments

Compare performance by customer segment, contract type, and usage pattern.

Next step: connect prediction to measurable intervention



Refresh

Validate the model on a refreshed customer snapshot.

Rank

Rank customers by predicted churn risk and expected value.

Test

Test retention actions with a controlled holdout group.

Monitor

Monitor lift, save rate, and customer experience over time.